

$$\begin{aligned}
& \wedge op_seq' = \\
& \quad [op_seq \text{ EXCEPT } \\
& \quad \quad ![t] = \\
& \quad \quad \quad Append(@, [type \mapsto \text{"W"}, key \mapsto op.key, val \mapsto op.val])] \\
& \wedge pc' = [pc \text{ EXCEPT } ![t] = pc[t] + 1] \\
& \wedge \text{UNCHANGED } \langle client_dep, \\
& \quad \quad client_last_commit, \\
& \quad \quad router_ct, \\
& \quad \quad shard_ct, \\
& \quad \quad tx_status, \\
& \quad \quad tx_snapshot, \\
& \quad \quad tx_commit_time, \\
& \quad \quad store \\
& \quad \rangle \\
CommitTx(t) & \triangleq \\
& \wedge tx_status[t] = \text{"Active"} \\
& \wedge pc[t] > Len(Workload[t].ops) \\
& \wedge \text{LET } IsConcurrent(t_other) \triangleq \\
& \quad tx_status[t_other] = \text{"Committed"} \wedge \\
& \quad tx_commit_time[t_other] > tx_snapshot[t] \\
HasConflict & \triangleq \\
& \exists t_other \in AllTxns : \\
& \quad \wedge IsConcurrent(t_other) \\
& \quad \wedge \exists k \in tx_checkset[t] : \\
& \quad \quad \exists v \in store[k] : \\
& \quad \quad \quad v.writer = t_other \wedge v.cts = tx_commit_time[t_other] \\
IN \quad IF HasConflict & \\
& \quad THEN \wedge tx_status' = [tx_status \text{ EXCEPT } ![t] = \text{"Aborted"}] \\
& \quad \wedge \text{UNCHANGED } \langle client_dep, \\
& \quad \quad client_last_commit, \\
& \quad \quad router_ct, \\
& \quad \quad shard_ct, \\
& \quad \quad tx_snapshot, \\
& \quad \quad tx_checkset, \\
& \quad \quad tx_buffer, \\
& \quad \quad tx_commit_time, \\
& \quad \quad pc, \\
& \quad \quad store, \\
& \quad \quad op_seq \\
& \quad \rangle \\
& \quad ELSE \wedge shard_ct' = shard_ct + 1 \\
& \quad \wedge tx_commit_time' = [tx_commit_time \text{ EXCEPT } ![t] = shard_ct'] \\
& \quad \wedge tx_status' = [tx_status \text{ EXCEPT } ![t] = \text{"Committed"}] \\
& \quad \wedge store' =
\end{aligned}$$